

Energy Consumption Analysis Using Statistical Estimation Techniques

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Introduction

Problem

Characterizing skewed regional power demand.

Approach

Statistics applied on real POSOCO records.

Why FDS?

Population-level inference — no ML/DL.

Methodology — Statistical Techniques

01 Descriptive Statistics

Mean, median, mode, variance & skewness across 16,599 records.

02 Sampling & Estimation

Bootstrap resampling (n=40, 2000 iters); 95% CI via t-distribution.

03 Correlation & Covariance

Pearson correlation across top 6 states reveals demand coupling.

04 Outlier Detection

Tukey 1.5×IQR method flags anomalous days per grid region.

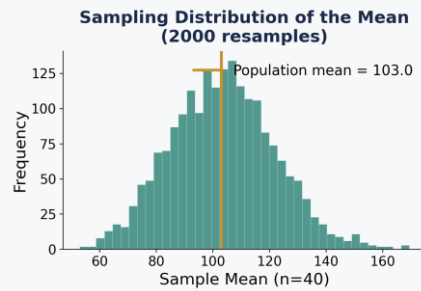
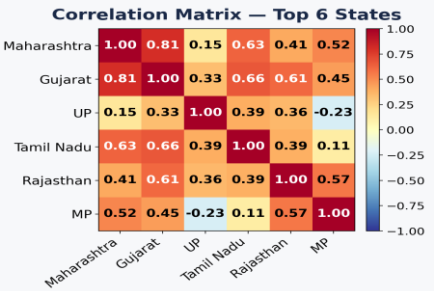
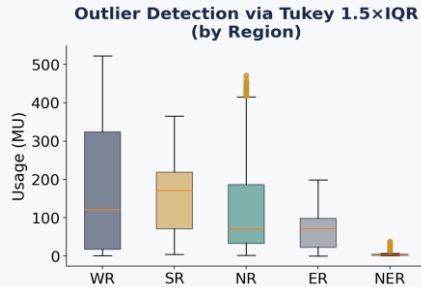
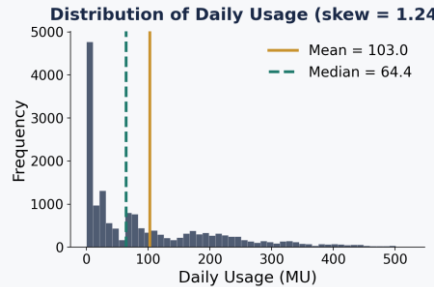
Conclusion & Future Directions

- Usage is right-skewed (skew = 1.24); WR & SR dominate demand, NER only ~6 MU/day.
- Tukey IQR flags 3.37% of records as anomalous, mostly within NER's narrow band.
- Maharashtra – Gujarat correlate strongly ($r = 0.807$, $p < 0.001$) via shared industrial demand.

Dataset — POSOCO State-wise Power Consumption

- Source: POSOCO, Ministry of Power, Govt. of India
- 16,600 records · Jan 2019–Dec 2020 · 5 regions, 29 states/UTs
- Fields: State, Region, Lat/Long, Date, Usage (MU)

Results — Key Findings



103.0
Mean

64.4
Median

116.0
Std Dev

3.37%
Outliers

0.807
Corr (MH-GJ)